STPM Physics · Semester 1 · Chapter 6

Gravitation

6.1 Newton's Law of Gravitation

Definitions & laws

- **Newton's law of gravitation**: every point mass in the universe attracts every other point mass with a force that is directly proportional to the product of their masses and inversely proportional to the square of the distance between them.
- The gravitational force is always attractive, acts along the line joining the two masses, and the two forces form an action–reaction pair (equal magnitude, opposite direction) on the two bodies.
- A uniform spherical body acts as if its mass were concentrated at its centre, so r is measured between centres.
- **Kepler's first law**: the orbit of a planet is an ellipse with the Sun at one focus.
- **Kepler's second law**: the line joining a planet to the Sun sweeps out equal areas in equal times.
- **Kepler's third law**: the square of the period of a planet is directly proportional to the cube of its mean orbital radius (semi-major axis), $T^2 \propto r^3$.
- **Weight** is the gravitational force exerted on a body by the Earth; $g = \frac{GM}{R^2}$ at the surface.

Formulae

Quantity / law	Formula	Symbols, units & conditions
Newton's law of gravitation	$F = \frac{GMm}{r^2}$	F = gravitational force (N); G = 6.67×10^{-11} N m^2 kg^{-2}; M, m = masses (kg); r = separation of the centres (m)
Surface gravitational field strength	$g = \frac{GM}{R^2}$; $GM = gR^2$	g = gravitational field strength at the surface (N kg^{-1}); M = mass of planet (kg); R = radius of planet (m)
Planet of density ρ	$M = \frac{4}{3}\pi R^3\rho$; $g = \frac{4}{3}\pi G\rho R$	M = mass (kg); R = radius (m); ρ = density (kg m^{-3}); G = gravitational constant
Circular orbit	$\frac{GMm}{r^2} = \frac{mv^2}{r} = mr\omega^2$; $T^2 = \frac{4\pi^2}{GM}r^3$	M = mass of central body (kg); m = orbiting mass (kg); r = orbital radius (m); v = orbital speed (m s^{-1}); ω = angular velocity (rad s^{-1}); T = period (s)
Kepler's third law	$\frac{T_1^2}{T_2^2} = \frac{r_1^3}{r_2^3}$	T_1, T_2 = periods (s); r_1, r_2 = orbital radii (m); two orbits about the same body
Mass of the central body	$M = \frac{4\pi^2r^3}{GT^2}$	M = mass of central body (kg); r = orbital radius (m); T = period (s); G = gravitational constant

Quantity / law	Formula	Symbols, units & conditions
Density method	$M = \rho \frac{4}{3}\pi R^3$; $g \propto R$	M = mass (kg); ρ = density (kg m^{-3}); R = radius (m); $g \propto R$ holds for equal density

How to answer (MPM style)

- State the law with "product of masses" and "square of the distance"; quote $F = \frac{GMm}{r^2}$ and define every symbol.
- In calculations convert km → m before squaring, and use r from the centre of the planet (R + h for a satellite at height h).
- For ratio questions write the ratio of formulae, cancel constants, then substitute (e.g. $\frac{F_2}{F_1} = \frac{m_2}{m_1}(\frac{r_1}{r_2})^2$).
- For Kepler's third law show $\frac{T^2}{r^3}$ is constant, then substitute; keep T and r in consistent units on both sides.

Section A traps

Trap: Forgotten square: doubling r reduces F to one quarter (not one half); halving r gives four times F.

Trap: Using the surface-to-surface distance or the height above the surface as r instead of the distance between centres.

Trap: Equal-density planets: $M \propto R^3$, so $g \propto R$ (not R^2 or $\frac{1}{R^2}$); $g \propto \frac{1}{R^2}$ applies only for a fixed mass.

Trap: Thinking the heavier body exerts the larger force; the two forces are equal in magnitude.

Tricks

Trick: Ratio method: $F \propto \frac{m_1 m_2}{r^2}$; write $F_2 = F_1 \times$ (mass factors) $\times$ (1/distance factor)^2.

Trick: Kepler III shortcut: $T \propto r^{3/2}$; if r is 4 times, T is 8 times; if r is 9 times, T is 27 times.

6.2 Gravitational Field and Potential

Definitions & laws

- **Gravitational field** is a region of space in which a mass experiences a gravitational force.
- **Gravitational field strength** g at a point is the gravitational force per unit mass acting on a small test mass placed at that point; g is a vector directed towards the mass producing the field.

- **Gravitational potential** V at a point is the work done per unit mass by an external agent in bringing a small test mass from infinity to that point; V = 0 at infinity, so V is negative everywhere in the field of an attractive mass, and it is a scalar.
- **Gravitational potential energy** U of a mass m at a point is the work done by an external agent in bringing the mass from infinity to that point, U = mV.
- Field strength is the negative of the potential gradient, $g = -\frac{dV}{dr}$; the area under a g–r graph gives −ΔV.
- **Escape speed** is the minimum speed at which a body must be projected from the surface of a planet so that it just reaches infinity, where its speed is zero.
- An equipotential surface is a surface on which all points have the same potential; no work is done moving a mass along it, and g is perpendicular to it.

Formulae

Quantity / law	Formula	Symbols, units & conditions
Gravitational field strength	$g = \frac{GM}{r^2}$	g = field strength (N kg^{-1}); M = mass of the source (kg); r = distance from centre (m); valid for r ≥ R (outside a spherical body)
Gravitational potential	$V = -\frac{GM}{r}$	V = gravitational potential (J kg^{-1}); M = mass of source (kg); r = distance from centre (m)
Gravitational potential energy	$U = -\frac{GMm}{r}$	U = potential energy (J); M = mass of source (kg); m = mass in the field (kg); r = distance from centre (m)
Field and potential gradient	$g = -\frac{dV}{dr}$	g = field strength (N kg^{-1}); V = potential (J kg^{-1}); r = distance (m)
Change in potential energy	$\Delta U = m(V_2 - V_1) = GMm(\frac{1}{r_1} - \frac{1}{r_2})$	ΔU = change in potential energy (J); V_1, V_2 = potentials (J kg^{-1}) at r_1, r_2 (m)
Change in potential energy near surface	$\Delta U = mg\Delta h$	m = mass (kg); g = gravitational field strength (N kg^{-1}); Δh = change in height (m); near the surface
Escape speed	$\frac{1}{2}mv^2 - \frac{GMm}{R} = 0$; $v_{esc} = \sqrt{\frac{2GM}{R}} = \sqrt{2gR}$	v_{esc} = escape speed (m s^{-1}); M = mass of planet (kg); R = radius (m); g = surface field strength (N kg^{-1}); for the Earth about 1.1×10^4 m s^{-1}
Energy conservation between two points	$\frac{1}{2}mv_1^2 - \frac{GMm}{r_1} = \frac{1}{2}mv_2^2 - \frac{GMm}{r_2}$	v_1, v_2 = speeds (m s^{-1}) at distances r_1, r_2 (m); m = mass (kg); M = mass of source (kg)
Several masses	$V = \Sigma V$	V = total potential (J kg^{-1}); scalar sum of the potentials due to each mass

- The field strength due to several masses is the vector sum of the individual fields.

How to answer (MPM style)

- Define g as "force per unit mass" and V as "work done per unit mass in bringing a small mass from infinity"; mention V = 0 at infinity.
- For two masses, add fields as vectors (direction by components) and potentials as scalars (with sign); the neutral point is where the resultant g is zero.
- For escape speed, equate the kinetic energy to the magnitude of the potential energy: $\frac{1}{2}mv^2 = \frac{GMm}{R}$; the mass of the projectile cancels.
- Sketch V against r: labelled axes, V negative, starting at $-\frac{GM}{R}$ at r = R and rising towards zero (never positive) as $r \to \infty$; g against r: positive magnitude decreasing as $\frac{1}{r^2}$.

Section A traps

Trap: V is always negative and increases towards zero as r increases; moving a mass away from the Earth increases its potential energy (U becomes less negative).

Trap: Using $\Delta U = mg\Delta h$ for large heights; it is valid only when g is nearly constant ($\Delta h \ll R$). Otherwise use $GMm(\frac{1}{r_1} - \frac{1}{r_2})$.

Trap: $V \propto \frac{1}{r}$ but $g \propto \frac{1}{r^2}$; doubling r halves V and quarters g. V is a scalar (J kg^{-1}); g is a vector (N kg^{-1}).

Trap: Escape speed does not depend on the mass or the direction of projection of the body, only on M and R of the planet; $v_{esc} = \sqrt{2}$ × orbital speed at the surface, not 2 ×.

Tricks

Trick: Unit check: J kg^{-1} = m^2 s^{-2} for V; N kg^{-1} = m s^{-2} for g.

Trick: Neutral point between two masses M_1 and M_2: $\frac{GM_1}{x^2} = \frac{GM_2}{(d-x)^2}$, i.e. the point lies closer to the smaller mass and $\frac{x}{d-x} = \sqrt{\frac{M_1}{M_2}}$.

6.3 Satellite Motion

Definitions & laws

- **Satellite** in a circular orbit is moving under the gravitational force of the central body, which provides the centripetal force: $\frac{GMm}{r^2} = \frac{mv^2}{r}$.

- **Geostationary satellite** is a satellite that stays above a fixed point on the Earth's equator: it orbits in the equatorial plane, in the same direction as the Earth's rotation (west to east), with period 24 h (same angular velocity as the Earth), at a fixed height of about 3.6×10^7 m above the surface ($r \approx 4.2\times10^7$ m).
- **Apparent weightlessness**: in orbit the satellite and its contents have the same acceleration towards the Earth (free fall), so there is no contact (normal) force between the astronaut and the floor, although the gravitational force still acts.
- **Kepler's third law**: $T^2 \propto r^3$ for orbits about the same central mass.
- The gravitational force on the satellite is perpendicular to its velocity in a circular orbit, so it does no work and the speed and total energy remain constant.
- A satellite in a lower orbit has a greater speed and a smaller period; to move to a higher orbit the satellite gains total energy (E becomes less negative).

Formulae

Quantity / law	Formula	Symbols, units & conditions
Orbital speed	$v = \sqrt{\frac{GM}{r}}$	v = orbital speed (m s^{-1}); M = mass of central body (kg); r = R + h = orbital radius (m)
Period and mass of central body	$T = \frac{2\pi r}{v}$; $T^2 = \frac{4\pi^2}{GM}r^3$; $M = \frac{4\pi^2 r^3}{GT^2}$	T = period (s); r = orbital radius (m); v = orbital speed (m s^{-1}); M = mass of central body (kg)
Angular velocity and acceleration	$\omega = \sqrt{\frac{GM}{r^3}}$; $a = g_r = \frac{GM}{r^2}$	ω = angular velocity (rad s^{-1}); a = centripetal acceleration = field strength g_r at radius r (m s^{-2})
Kinetic energy of satellite	$K = \frac{GMm}{2r}$	K = kinetic energy (J); m = satellite mass (kg); M = mass of central body (kg); r = orbital radius (m)
Potential energy of satellite	$U = -\frac{GMm}{r}$	U = potential energy (J); m = satellite mass (kg); M = mass of central body (kg); r = orbital radius (m)
Total energy of satellite	$E = K + U = -\frac{GMm}{2r}$; $E = -K$; $U = -2K$	E = total energy (J); K = kinetic energy (J); U = potential energy (J)
Orbital speed near the surface	$v = \sqrt{gR} \approx 7.9\times10^3$ m s^{-1}	v = orbital speed (m s^{-1}); g = surface field strength (N kg^{-1}); R = radius of planet (m)
Escape speed from orbital speed	$v_{esc} = \sqrt{2}\, v_{orbit}$	v_{esc} = escape speed (m s^{-1}); v_{orbit} = orbital speed (m s^{-1})
Energy to raise a satellite	$\Delta E = \frac{GMm}{2}\left(\frac{1}{r_1} - \frac{1}{r_2}\right)$	ΔE = energy needed (J); r_1, r_2 = initial and final orbital radii (m); m = satellite mass (kg); M = mass of central body (kg)
Geostationary radius	$r^3 = \frac{GMT^2}{4\pi^2}$	r = orbital radius (m); T = 24 h = 8.64×10^4 s; M = mass of Earth (kg)

How to answer (MPM style)

- Start with "gravitational force provides the centripetal force" and equate $\frac{GMm}{r^2}$ to $\frac{mv^2}{r}$ (or $mr\omega^2$); the satellite mass cancels.
- Always use $r = R + h$ in the formulas; substitute M and G or use $GM = gR^2$ when g and R are given.
- State the three conditions for geostationary orbit: equatorial plane, period 24 h, same direction as the Earth's spin (west to east).
- For weightlessness, explain with free fall: satellite and astronaut have the same acceleration, so the reaction force is zero; do not say gravity is zero.

Section A traps

Trap: "Weightless" does not mean zero gravity: g at orbital height is still large (about 8–9 N kg^{-1} at 400 km).

Trap: A higher orbit has a slower speed and longer period; the speed does not increase with r.

Trap: The total energy of a bound satellite is negative, $E = -\frac{GMm}{2r}$; do not use $+\frac{GMm}{2r}$, and do not take E as equal to K.

Trap: Orbital speed and period do not depend on the satellite mass; a heavier satellite at the same r has the same v and T.

Trap: Using the height h above the surface in place of $r = R + h$ in $v = \sqrt{\frac{GM}{r}}$ or Kepler's law.

Tricks

Trick: Ratio method: $v \propto r^{-1/2}$, $T \propto r^{3/2}$, $\omega \propto r^{-3/2}$; doubling r reduces v by $\frac{1}{\sqrt{2}}$.

Trick: Energy shortcut: $K = |E|$ and $U = 2E$, so if r doubles, K and $|E|$ halve.